



STRUCTURED QUERY LANGUAGE

Cheat Sheet (SQL Basics)

SQL Data Types

Exact Numeric's:

- INTEGER
- SMALLINT
- BIGINT
- NUMERIC
- DECIMAL

Binary Strings:

- BINARY
- BINARY VARYING
- BINARY LARGE OBJECT

Character Strings:

- CHARACTER
- CHARACTER VARYING (VARCHAR)
- CHARACTER LARGE OBJECT
- NATIONAL CHARACTER
- NATIONAL CHARACTER VARYING
- NATIONAL CHARACTER LARGE OBJECT

Collection Types:

- ARRAY
- MULTI

Approximate Numeric's:

- REAL
- DOUBLE PRECISION
- FLOAT
- DECFLOAT

Boolean:

Intervals:

- INTERVAL DAY
- INTERVAL YEAR

Date Times:

- DATE
- TIME WITHOUT TIMEZONE
- TIMESTAMP WITHOUT TIMEZONE
- TIME WITH TIMEZONE
- TIMESTAMP WITH TIMEZONE

Other Types:

- ROW
- XML

View

It is a virtual table which is a result of a query. It is often used as a security mechanism letting users to access the data through the views.

Syntax:

```
CREATE VIEW view1 AS
SELECT 1,2
FROM 1
WHERE condition
```

Indexes

It is used to speed up the performance of the queries by reducing the number of database pages to be visited.

Syntax:

- To create an Index:
CREATE INDEX index_name ON t(c1, c2)
- To create an unique index:
CREATE UNIQUE INDEX index_name ON t(c3, c4)
- To drop an index:
DROP INDEX Index_name

Function

Aggregate Functions: It is a function where the values of multiple rows are combined to form a single value.

UNION: A set operation can be used on the returned results called 'UNION' which can append the result of one query to another.

Syntax:

```
SELECT Col1, Col2 FROM Table1
UNION
SELECT Col3, Col4 FROM Table2
```

Stored Procedure

It is a set of SQL statements with assigned names that can be shared and reused by multiple programs

Syntax:

To create Procedure

```
CREATE PROCEDURE procedure_name
@variable AS datatype = value
AS
-Comments
SELECT*FROM t GO
```

Using SQL Constraints

1. PRIMARY KEY:

Set c1 and c2 as primary key

Syntax:

```
CREATE TABLE t(  
  c1 INT, c2 INT, c3 VARCHAR,  
  PRIMARY KEY (c1,c2)  
);
```

2. FOREIGN KEY:

Set c2 column as a foreign key

Syntax:

```
CREATE TABLE t1(  
  c1 INT PRIMARY KEY,  
  c2 INT,  
  FOREIGN KEY (c2) REFERENCES t2(c2)  
);
```

Unique

Making the values in C1 and C2 as unique.

Syntax:

```
CREATE TABLE t(  
  c1 INT, c1 INT,  
  UNIQUE (c2,c3)  
);
```

Trigger

It is a special type of stored procedure that automatically executes when a user tries to modify through a DML event.

Syntax:

- To create or modify trigger

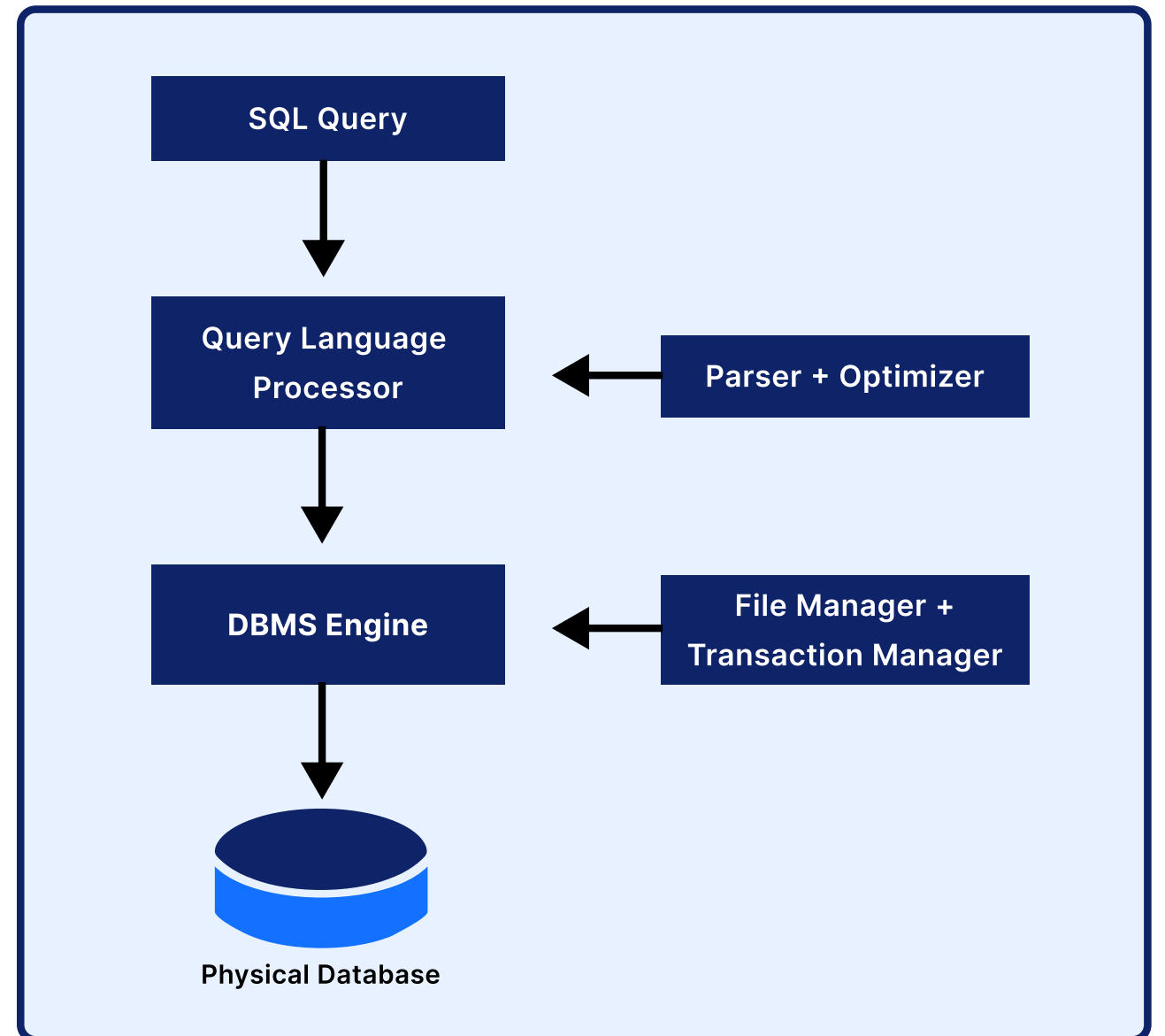
```
CREATE OR MODIFY TRIGGER  
  trigger_name  
  WHEN EVENT  
  ON table_name TRIGGER_TYPE  
  EXECUTE stored_procedure
```

- To delete or drop a trigger: Used to delete a specific trigger.

```
DROP TRIGGER trigger_name
```

Explanation

- WHEN;
 - i) BEFORE: Invokes before an event occurs
 - ii) AFTER: Invokes after an event occurs
- EVENT:
 - i) INSERT: Invokes for Insert
 - ii) UPDATE: Invokes for Update
 - iii) DELETE: Invoke for Delete
- TRIGGER TYPE:
 - i) FOR EACH ROW
 - ii) FOR EACH STATEMENT



FUNCTION	DESCRIPTION
TO_DATE	It is used to convert a string to date.
COALESCE	Returns the first non NULL results, when querying with the columns that contain NULL.
CURRENT_TIME STAMP	Returns the correct time on the database server.
COUNT	An aggregate function that returns the number of rows in the result set.
SUM	An aggregate function that sums up the values in a result set.
AVG	To compute the mean average of the values in the results set.
MIN / MAX	An aggregate function to return the largest/smallest value among the result set.
LISTAGG	It is used to transform values from a group of rows into delimited string.

OPERATOR	SYNTAX	DESCRIPTION
UNION	SELECT C1 FROM t1 UNION [ALL] SELECT C1 FROM t2	Select column 1 from table t1 and column 2 from table t2 and combine rows of these queries.
INTERSECT	SELECT C1 FROM t1 INTERSECT SELECT C1 FROM t2	It is used to return the intersection of two queries.
MINUS	SELECT C1 FROM t1 MINUS SELECT C1 FROM t2	It is used to subtract the second result set from the first.
NOT LIKE	SELECT c1 FROM t WHERE c1 [NOT] LIKE pattern	It is used to return the query of rows using the matching pattern.
BETWEEN	SELECT c1 FROM t WHERE c1 BETWEEN min AND max	It returns the rows where c1 is between MIN and MAX
NOT NULL	SELECT c1 FROM t WHERE c1 IS [NOT] NULL	To check if the values are NULL or NOT NULL.